

# Oscar Johan Høeg Wohlfahrt

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## PROFILE

MSc student in Computational Science and Engineering at EPFL with interests spanning machine learning, optimization, scientific computing, and AI for scientific discovery. Experience across ML/LLM research, software engineering, and data-intensive projects, with a particular interest in technically challenging problems that combine strong quantitative reasoning with building systems that work in practice.

## EDUCATION

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| <b>École Polytechnique Fédérale de Lausanne (EPFL)</b><br><i>MSc Computational Science and Engineering</i><br><i>Selected coursework:</i> Algorithms II, High Performance Computing, Reinforcement Learning, Mathematics of Data, Distributed Algorithms, Scientific Computing, Stochastic Simulation, Computational Neuroscience | <b>Sep. 2025 – Jun. 2027</b><br>Lausanne, Switzerland |
| <b>Technical University of Denmark (DTU)</b><br><i>BSc General Engineering, Computer Science</i><br><i>Selected coursework:</i> Machine Learning & Data Mining, Algorithms & Data Structures, Numerical Algorithms, Function Spaces & Mathematical Analysis, Statistics, Database Systems                                         | <b>Aug. 2022 – Jun. 2025</b><br>Copenhagen, Denmark   |
| <b>New York University (NYU) – Exchange Semester</b><br><i>Selected coursework:</i> Applied Probability, Advanced Linear Algebra, Mathematical Modeling                                                                                                                                                                           | <b>Aug. 2024 – Dec. 2024</b>                          |
| <b>Università Cattolica del Sacro Cuore</b><br><i>BSc Banking and Finance – Completed first year</i><br>Completed the first year before changing academic direction toward engineering and computer science.                                                                                                                      | <b>Aug. 2021 – Jun. 2022</b><br>Milan, Italy          |

## EXPERIENCE

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| <b>Accenture – Telenor AI Factory</b><br><i>Software Engineering Intern</i> <ul style="list-style-type: none"><li>Built full-stack features for Telenor AI Factory’s Model-as-a-Service portal, including model access, API-key management, usage and billing analytics, and organization/team administration.</li><li>Developed Kotlin/Spring Boot services with PostgreSQL and LiteLLM integrations alongside a React/TypeScript frontend, including authentication, authorization, error handling, and automated tests.</li></ul> | <b>Jun. 2026 – Aug. 2026</b><br>Oslo, Norway        |
| <b>Tet Digital / Ruter AI Lab</b><br><i>AI / ML Intern</i> <ul style="list-style-type: none"><li>Developed an LLM-based data agent for RuterGPT, enabling natural-language access to internal Snowflake data and generating SQL, analyses, and visualizations.</li><li>Helped redesign a complex RAG/text-to-SQL workflow into a modular tool-based agent, improving reliability and maintainability through predefined queries, caching, and output compression.</li></ul>                                                          | <b>Jun. 2025 – Aug. 2025</b><br>Oslo, Norway        |
| <b>L-Shop-Team</b><br><i>Student Assistant</i> <ul style="list-style-type: none"><li>Worked with database management, proofreading, and customer communication in a multilingual environment.</li></ul>                                                                                                                                                                                                                                                                                                                              | <b>Oct. 2023 – Jul. 2024</b><br>Copenhagen, Denmark |
| <b>Sult</b><br><i>Event Manager / Waiter</i> <ul style="list-style-type: none"><li>Managed and worked at weddings, corporate events, and other large events, coordinating preparation, service, and practical execution.</li></ul>                                                                                                                                                                                                                                                                                                   | <b>Feb. 2018 – Aug. 2023</b><br>Oslo, Norway        |
| <b>Norske Studenter Roklubb</b><br><i>Rowing Coach</i> <ul style="list-style-type: none"><li>Coached junior rowers aged 12–15 through weekly training sessions and summer camps.</li></ul>                                                                                                                                                                                                                                                                                                                                           | <b>May 2020 – Aug. 2023</b><br>Oslo, Norway         |

## RESEARCH & PROJECTS

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| <b>Adaptive Inference-Time Compute Allocation for Lean Proof Generation</b><br><i>EPFL – Chair of Mathematical Data Science – Supervised by Etienne Bamas</i> <ul style="list-style-type: none"><li>Implemented and evaluated inference-time sampling strategies for LLM-based Lean proof generation using Apertus-8B and DeepSeek-Prover-V2; investigated adaptive compute allocation and model uncertainty.</li></ul> | <b>Spring 2026</b>         |
| <b>Automated Multi-Hop QA Generation for LLM Post-Training</b><br><i>Swiss AI Initiative, EPFL – Supervised by Eduard Dürsch and Valentina Pyatkin</i> <ul style="list-style-type: none"><li>Designed and implemented an end-to-end pipeline for generating document-grounded multi-hop QA data from long-context documents for downstream LLM post-training.</li></ul>                                                 | <b>Spring 2026</b>         |
| <b>Test-Time Adaptation for RNA Language Models</b>                                                                                                                                                                                                                                                                                                                                                                     | <b>Fall 2026 – Ongoing</b> |

*EPFL — Signal Processing Laboratory 4 (LTS4) — Supervised by Vincent Jung*

- Investigating test-time adaptation for pretrained RNA language models, initially focusing on zero-shot RNA structure prediction under distribution shift.

### **Machine Learning for Building Lifetime Modeling**

**Spring 2025**

*DTU Compute — Bachelor's Thesis — Supervised by Nicolai Siim Larsen*

- Worked with large-scale Danish building data and compared classical, ensemble, and deep survival-analysis approaches, including extensive data cleaning and feature analysis.

### **TECHNICAL SKILLS**

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**Languages:** Python, C++, Java, SQL, Kotlin, JavaScript/TypeScript, F#

**ML / Scientific:** PyTorch, scikit-learn, NumPy, SciPy, Hugging Face

**HPC / Systems:** CUDA, MPI, OpenMP, SLURM, Docker, Linux, AWS, PostgreSQL

### **ADDITIONAL**

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**Norwegian National Rowing Team** — Junior / U23 athlete; competed nationally and internationally

2018 – 2021